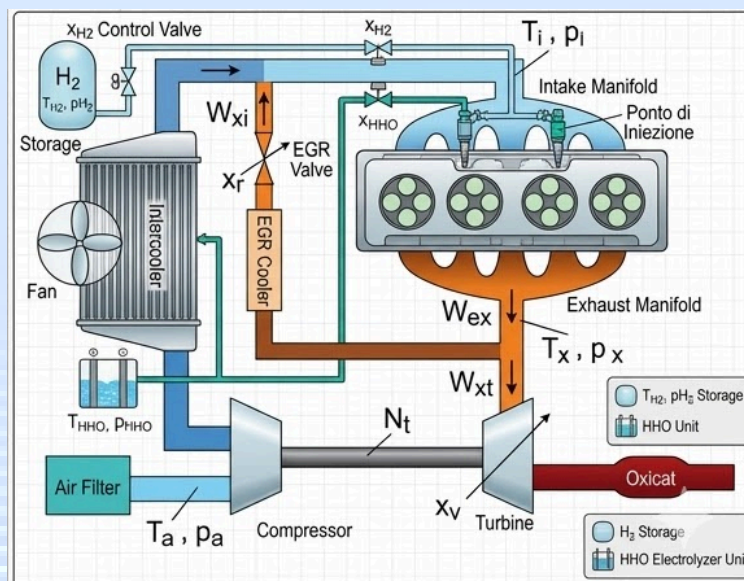
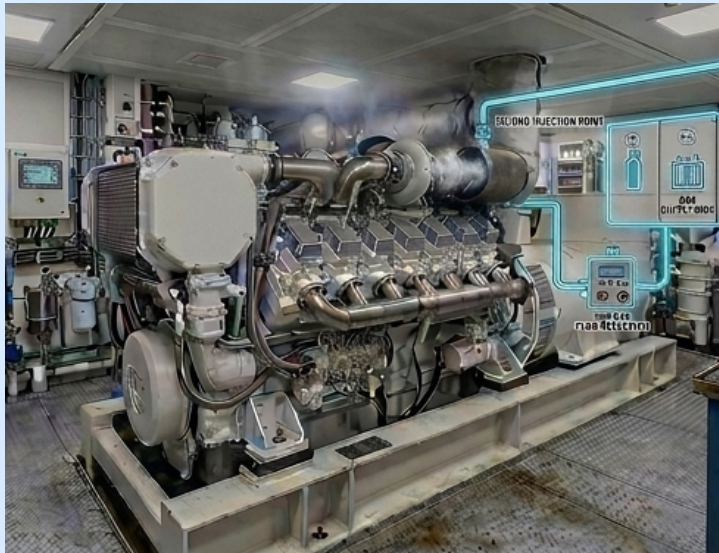


Combustion air enrichment in diesel engines using hydrogen and oxyhydrogen



Comparative analysis of combustion, performance, emissions, and industrial feasibility

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Abstract

The enrichment of the combustion air of diesel engines with gases produced by electrolysis has been the subject of numerous experimental studies over the last two decades. In particular, the use of **separated hydrogen (H_2)** and **oxyhydrogen ($H_2 + O_2$)** showed different effects on combustion, consumption and emissions.

This work critically analyses the two approaches applied to **large diesel engines**, with reference to a stationary group of ~ 2.2 MW, integrating **energy calculations, experimental data from literature and safety and standardization assessments**.

1. Technical context

Stationary and marine diesel engines have specific characteristics:

- operation at near constant speed;
- very lean air/fuel ratio ($\lambda \gg 1$);
- high-pressure common-rail diesel injection;
- economic relevance of even small improvements (%) in efficiency.

In this context, hydrogen **it is not considered as the main fuel**, but how **combustion additive**, introduced into the suction.

2. Analysis methodology

The analysis combines:

1. **Energy balances**
2. **Expected benefit modeling**
3. **Experimental results published**
4. **Industrial risk assessment**

Benchmarks include:

- specific fuel consumption;
- combustion development (heat release);
- PM and NO_x emissions;
- system complexity.

3. Enrichment with hydrogen (H₂)

3.1 Combustion mechanism

Hydrogen has:

- flame speed 5–7 times higher than diesel;
- high molecular diffusivity;
- very low ignition energy.

In diesel:

- anticipates and stabilizes auto-ignition;
- reduces locally rich pockets;
- improves the premixed combustion fraction.

These effects are widely documented in the literature on **dual-fuel hydrogen-diesel engines**. [\[link.springer.com\]](https://link.springer.com), [\[mdpi.com\]](https://mdpi.com)

3.2 Dimensional calculations (case study 2.19MW)

Estimated diesel consumption

$$\text{BSFC} \approx 200 \text{ g/kWh} \Rightarrow 438 \text{ kg/h}$$

Input energy

$$438 \times 42 = 18,4 \text{ GJ}$$

Effective H₂ share (literature): 0.5–3% energy

H ₂ energy share	H ₂ required
0,5%	~0.8 kg/h
1,5%	~2.3 kg/h
3%	~4.6 kg/h

PEM electrolyzer power (55 kWh/kg):

- ~50 kW → 250 kW

Values consistent with existing industrial systems. [\[mdpi.com\]](https://www.mdpi.com),
[\[sites.ualberta.ca\]](https://sites.ualberta.ca)

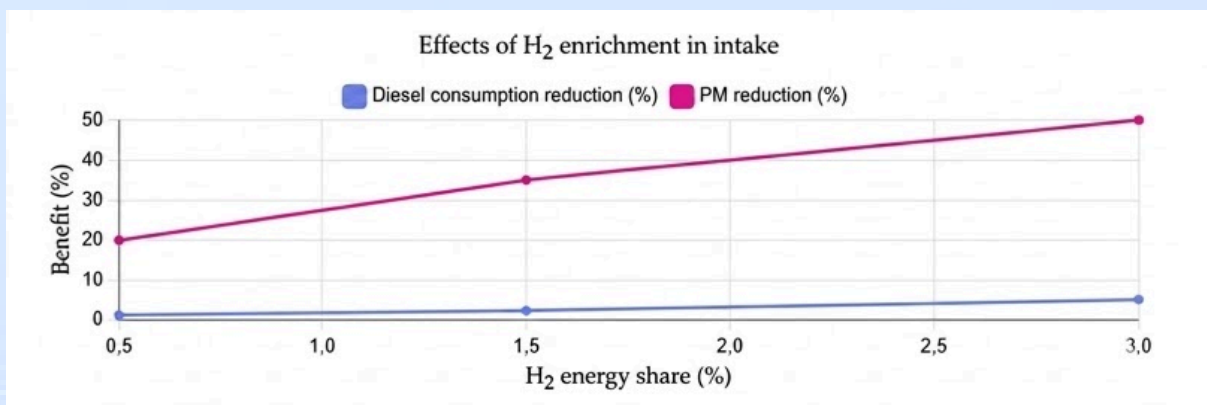
3.3 Measured Benefits

Summary of experimental results reported in the literature:

- **PM:** -20% → -50% [pubs.rsc.org], [sites.ualberta.ca]
- **Diesel consumption:** -1% → -4% [mdpi.com], [international...alssrg.org]
- **Combustion stability:** improved at partial loads [mdpi.com]

3.4 Benefits summary chart

The following graph summarizes the expected trend as a function of the energy share of H₂:



Key observation:

the benefit on PM is more marked than the fuel savings, making H₂ particularly interesting as a **emission control instrument**.

4. Enrichment with oxyhydrogen ($\text{H}_2 + \text{O}_2$)

4.1 Principle

Oxyhydrogen is a nearly stoichiometric H_2/O_2 mixture, already detonating, introduced directly into the intake air.

4.2 Observed effects

Some studies show:

- PM reduction (up to -60%);
- slight reduction in CO and HC;
- moderate increase in NO_x ($+5 \rightarrow +15\%$). [\[pubs.rsc.org\]](https://pubs.rsc.org)

4.3 Structural limitations

However, on an industrial scale, significant problems arise:

- Oxygen **it is not limiting** in modern diesels;
- excessively rapid combustion;
- increase in $\text{dp}/\text{d}\theta$ and mechanical stress;
- **extremely high ATEX risk.**

These aspects make oxyhydrogen **non-scalable** on multi-MW motors, as confirmed by the lack of consolidated industrial applications.

[\[CodeInterpreter | InteractiveChart\]](#)

5. Comparison of H₂ vs. oxyhydrogen

Topic	H ₂ separated	Oxyhydrogen
Combustion control	High	Low
PM	↓↓↓	↓↓
NO _x	Manageable	Critical
Safety	Manageable	Problem
Normability	Good	Poor
Scalability	Industrial	Limited

6. System implications

Only the solution **H₂ separated** allows:

- fine dosing (MFC);
- quick stop logic;
- ATEX certifiability;
- integration with EGR and engine control.

Oxyhydrogen remains confined to laboratory studies or small experimental engines.

7. Economic analysis of hydrogen enrichment in aspiration

7.1 Objective of the analysis

Evaluate the **economic convenience** of the integration of a system for the production and use of hydrogen (H₂) in aspiration on a **2.19MW stationary diesel engine** considering:

- reduction of diesel consumption;
 - reduction of environmental/emission costs (where valorisable);
 - investment costs (CAPEX);
 - operating costs (OPEX);
 - payback times.
-

7.2 Basic data (case study)

Motor

- Rated electrical power: **2 190 kW**
- Annual hours of operation: **6 000 h/a**
(typical for continuous service stationary/naval groups)
- Diesel consumption:

438 kg/h ≈ 525 L/h

Diesel price (industrial, European average)

- **1,00 €/L** (conservative assumption 2025–2026)

Annual diesel expenditure (baseline)

$$525 \text{ L/h} \times 6\,000 \text{ h} = 3\,150\,000 \text{ L/a}$$

$$\Rightarrow 3,15 \text{ M€/a}$$

7.3 H₂ operating scenario (selected for analysis)

For the economic analysis we choose the **nominal scenario**, technically more realistic:

- hydrogen = **1.5% of diesel energy**
- H₂ required: **2.3 kg/h**
- PEM electrolyzer: **250 kW**
- reduction in diesel consumption: **2%**
- PM reduction: **~35%**

7.4 Direct economic benefits

7.4.1 Diesel savings

Reducing consumption:

$$3,15 \text{ M€/a} \times 0,02 = 63\,000 \text{ €/a}$$

This entry alone does not justify the investment.

7.4.2 Environmental enhancement (when applicable)

It depends on the context (ETS, port emissions, ESG KPIs).

Prudent case:

- Alone **PM and maintenance**
- Encrustation reduction, filters, DPF regenerations, system shutdown

Typical estimable value:

- **10–30 k€/a**

Let's assume an average value:

20 000 €/a

7.4.3 Overall operating benefit

$63\,000 + 20\,000 = \mathbf{83\,000\ €/a}$

7.5 Operating Expenses (OPEX)

7.5.1 Electricity from electrolysis

H₂ required:

$2,3\text{ kg/h} \times 55\text{ kWh/kg} = 127\text{ kW}$

On an annual basis:

$$127 \times 6\,000 = 762\,000 \text{ kWh/a}$$

Case A – “surplus” renewable electricity

- marginal cost $\approx 0 \text{ €/kWh}$
- OPEX ≈ 0

✓ Ideal case (dedicated PV/wind systems, microgrids)

Case B – electricity from industrial grid

- average cost: $0,12 \text{ €/kWh}$

$$\Rightarrow 91\,400 \text{ €/a}$$

⚠ In this case the system it is not economically convenient.

7.6 Estimated CAPEX (order of magnitude)

Voice	Approximate cost
PEM electrolyzer 250 kW	220–280 k€
H ₂ treatment (dryer, filters)	40–60 k€
H ₂ Buffer + Security	40–70 k€
Regulation, MFC, sensors	30–50 k€
Integration, piping, ATEX	40–60 k€

Estimated total CAPEX

370–520 k€

Let's take central value:

450 k€

7.7 Payback time

✓ Economically **SENSIBLE** scenario

(renewable electricity available)

$\text{Payback} = 450\,000 / 83\,000 \approx 5,4 \text{ years}$

→ acceptable for industrial plant

✗ **NON-SENSITIVE SCENARIO**

(electricity purchased from the grid)

Annual balance = €83,000 – €91,400 ≈ – €8,400/year

→ infinite payback

7.8 Economic Sensitivity (What Changes Everything)

The system becomes interesting if **at least one** of these factors it is true:

1. ☒ marginal cost renewable electricity ≈ 0
2. ☒ diesel cost $> 1,20 \text{ €/L}$
3. ☒ CO₂ valorization / ETS / port emissions
4. ☒ use $> 7\,000 \text{ h/a}$
5. ☒ primary need for PM reduction (regulation)

7.9 Economic comparison of H₂ vs. oxyhydrogen

Voice	H ₂ separated	Oxygen
CAPEX	High	Half
OPEX	Optimizable	Invalid
Payback	Possible	Uncertain
Reliability	Industrial	Low
Insurability	☒	☒
Value over time	☒	☒

👉 Also if HHO were cheaper, would not overcome regulatory and risk barriers, therefore it does not create real value.

8. Final economic conclusion

👉 Hydrogen in aspiration is NOT a purely economic solution, is a technical-environmental solution that becomes economical only in certain contexts.

It is convenient when:

- ☒ you have renewables available
- ☒ the engine works for many hours

- ✓ emissions are a real constraint
- ✓ Diesel is expensive

It is not convenient when:

- ✗ electricity is bought
- ✗ the goal is only to save fuel

9 . Conclusions

The enrichment of the combustion air with **separated hydrogen**, produced by electrolysis powered by renewables:

- represents **areal optimization solution** for large diesel engines;
- offers concrete benefits on particulate emissions;
- allows for a moderate but economically significant reduction in consumption;
- it is consistent with progressive decarbonisation strategies.

On the contrary, the use of **oxyhydrogen**:

- has limited advantages;
- It introduces safety and control issues that are difficult to accept in an industrial setting.